

**CS536 Science of Programming**  
**Spring 2025**  
**Assignment 1 Sample Solution Sketches**

1. (a)  $(p \rightarrow (p \vee q)) \rightarrow q \equiv (p \rightarrow p \vee q) \rightarrow q$   
 (b)  $((p \leftrightarrow q) \rightarrow (r \leftrightarrow q)) \equiv (p \leftrightarrow q) \rightarrow (r \leftrightarrow q)$   
 (c)  $(p \vee ((q \wedge (r \vee p)) \wedge r)) \equiv p \vee q \wedge (r \vee p) \wedge r$   
 (d)  $\forall x \in \mathbb{Z}. ((x > 0) \rightarrow (\exists y. (y < x))) \equiv \forall x \in \mathbb{Z}. x > 0 \rightarrow \exists y. y < x$   
 (e)  $(\forall x > 0. (\forall y > 0. (x * y = 1))) \rightarrow ((x = y) \wedge (x = 1)) \equiv (\forall x > 0. \forall y > 0. x * y = 1) \rightarrow x = y \wedge x = 1$

2. (a) In the following truth table, we can see that  $(p \vee q) \wedge q$  and  $q$  always have the same truth value, so they are logically equivalent to each other.

| $p$ | $q$ | $p \vee q$ | $(p \vee q) \wedge q$ |
|-----|-----|------------|-----------------------|
| $T$ | $T$ | $T$        | $T$                   |
| $T$ | $F$ | $T$        | $F$                   |
| $F$ | $T$ | $T$        | $T$                   |
| $F$ | $F$ | $F$        | $F$                   |

- (b) In the following truth table, we can see that  $\neg(p \leftrightarrow q)$  and  $\neg p \leftrightarrow q$  always have the same truth value, so they are logically equivalent to each other.

| $p$ | $q$ | $p \leftrightarrow q$ | $\neg(p \leftrightarrow q)$ | $\neg p$ | $\neg p \leftrightarrow q$ |
|-----|-----|-----------------------|-----------------------------|----------|----------------------------|
| $T$ | $T$ | $T$                   | $F$                         | $F$      | $F$                        |
| $T$ | $F$ | $F$                   | $T$                         | $F$      | $T$                        |
| $F$ | $T$ | $F$                   | $T$                         | $T$      | $T$                        |
| $F$ | $F$ | $T$                   | $F$                         | $T$      | $F$                        |

- (c) In the following truth table, we can see the column that represents the truth value of  $\neg p \wedge (p \vee q) \rightarrow q$  is always true, so  $\neg p \wedge (p \vee q) \rightarrow q$  is logically equivalent to  $T$ .

| $p$ | $q$ | $\neg p$ | $p \vee q$ | $\neg p \wedge (p \vee q)$ | $\neg p \wedge (p \vee q) \rightarrow q$ |
|-----|-----|----------|------------|----------------------------|------------------------------------------|
| $T$ | $T$ | $F$      | $T$        | $F$                        | $T$                                      |
| $T$ | $F$ | $F$      | $T$        | $F$                        | $T$                                      |
| $F$ | $T$ | $T$      | $T$        | $T$                        | $T$                                      |
| $F$ | $F$ | $T$      | $F$        | $F$                        | $T$                                      |

3. (a) Here I start from the left-hand side of this equality.

$$\begin{aligned}
 (p \vee q) \wedge \neg q &\Leftrightarrow (p \wedge \neg q) \vee (q \wedge \neg q) && \text{Distributivity of } \wedge \\
 &\Leftrightarrow (p \wedge \neg q) \vee F && \text{Contradiction} \\
 &\Leftrightarrow p \wedge \neg q && \text{Identity} \\
 &\Leftrightarrow \neg(p \rightarrow q) && \text{Negation of Implication}
 \end{aligned}$$

(b) Here I start from the right-hand side of this equality.

$$\begin{aligned}
\neg q \vee q \wedge \neg p &\Leftrightarrow (\neg q \vee q) \wedge (\neg q \vee \neg p) && \text{Distributivity of } \vee \\
&\Leftrightarrow T \wedge (\neg q \vee \neg p) && \text{Excluded Middle} \\
&\Leftrightarrow \neg q \vee \neg p && \text{Identity} \\
&\Leftrightarrow \neg(p \wedge q) && \text{DeMorgan's Law}
\end{aligned}$$

(c) Here I start from the right-hand side of this equality.

$$\begin{aligned}
(p \rightarrow q) \wedge (\neg p \rightarrow q) &\Leftrightarrow (\neg p \vee q) \wedge (\neg p \rightarrow q) && \text{Definition of } \rightarrow \\
&\Leftrightarrow (\neg p \vee q) \wedge (p \vee q) && \text{Definition of } \rightarrow \text{ and Double Negation} \\
&\Leftrightarrow (\neg p \wedge p) \vee q && \text{Distributivity of } \vee \\
&\Leftrightarrow q && \text{Contradiction and Identity}
\end{aligned}$$

4. (a)

$$\begin{aligned}
\neg(p \wedge q) \wedge p &\Leftrightarrow (\neg p \vee \neg q) \wedge p && \text{DeMorgan's Law} \\
&\Leftrightarrow (\neg p \wedge p) \vee (\neg q \wedge p) && \text{Distributivity of } \wedge \\
&\Leftrightarrow \neg q \wedge p && \text{Contradiction and Identity} \\
&\Rightarrow \neg q && \text{and-elimination}
\end{aligned}$$

(b)

$$\begin{aligned}
p \wedge q \vee q \wedge r &\Rightarrow p \vee q \wedge r && \text{and-elimination} \\
&\Rightarrow p \vee q && \text{and-elimination} \\
&\Rightarrow p \vee q \vee r && \text{or-introduction}
\end{aligned}$$

(c)

$$\begin{aligned}
(p \rightarrow q) \wedge (\neg p \rightarrow r) &\Leftrightarrow T \wedge (p \rightarrow q) \wedge (\neg p \rightarrow r) && \text{Identity} \\
&\Leftrightarrow (p \vee \neg p) \wedge (p \rightarrow q) \wedge (\neg p \rightarrow r) && \text{Excluded Middle} \\
&\Leftrightarrow p \wedge (p \rightarrow q) \wedge (\neg p \rightarrow r) \vee \\
&\quad \neg p \wedge (p \rightarrow q) \wedge (\neg p \rightarrow r) && \text{Distributivity of } \wedge \\
&\Rightarrow q \wedge (\neg p \rightarrow r) \vee \\
&\quad \neg p \wedge (p \rightarrow q) \wedge (\neg p \rightarrow r) && \text{Modus Ponens} \\
&\Rightarrow q \wedge (\neg p \rightarrow r) \vee \\
&\quad r \wedge (p \rightarrow q) && \text{Modus Ponens} \\
&\Rightarrow q \vee r \wedge (p \rightarrow q) && \text{and-elimination} \\
&\Rightarrow q \vee r && \text{and-elimination}
\end{aligned}$$

5. (a) True. With the given state,  $x > 2$  is evaluated to False and  $p$  is evaluated to True; so  $x > 2 \vee q$  is evaluated to True.

(b) False. Variable  $b$  has different data-types in the predicate and in the state.

(c) True.  $\sigma(p \leftrightarrow (\sigma(b[b[0]]) = 2)) = (\sigma(p) \leftrightarrow (\sigma(b[2]) = 2)) = (F \leftrightarrow F) = T$ .

(d) False. The proposition  $p \rightarrow p \vee q$  is tautology (proved by the or-introduction rule).

(e) True. The proposition/proposition  $T$  is valid so an empty state satisfies it.

6. (a)

$$\begin{aligned}
& \neg \forall x. x \geq 1. x^2 > x \\
& \equiv \neg \forall x. x \geq 1 \rightarrow x^2 > x && \text{Definition of bounded quantifier} \\
& \Leftrightarrow \exists x. \neg(x \geq 1 \rightarrow x^2 > x) && \text{DeMorgan's Law} \\
& \Leftrightarrow \exists x. x \geq 1 \wedge x^2 \leq x && \text{Negation of } \rightarrow
\end{aligned}$$

(b)

$$\begin{aligned}
& \neg \exists x. \exists y. x > y \wedge x < y \\
& \Leftrightarrow \forall x. \neg \exists y. x > y \wedge x < y && \text{DeMorgan's Law} \\
& \Leftrightarrow \forall x. \forall y. \neg(x > y \wedge x < y) && \text{DeMorgan's Law} \\
& \Leftrightarrow \forall x. \forall y. x \leq y \vee x \geq y && \text{DeMorgan's Law}
\end{aligned}$$

(c)

$$\begin{aligned}
& \neg \left( (\exists x. \exists y. Q(x, y)) \wedge \forall x. \forall y. Q(y, x) \right) \\
& \Leftrightarrow \neg (\exists x. \exists y. Q(x, y)) \vee \neg \forall x. \forall y. Q(y, x) && \text{DeMorgan's Law} \\
& \Leftrightarrow (\forall x. \neg \exists y. Q(x, y)) \vee \exists x. \neg \forall y. Q(y, x) && \text{DeMorgan's Law} \\
& \Leftrightarrow (\forall x. \forall y. \neg Q(x, y)) \vee \exists x. \exists y. \neg Q(y, x) && \text{DeMorgan's Law}
\end{aligned}$$

7. There are more than one correct ways to define these predicate functions.

(a)  $isGreater(b, m, x) \equiv 0 < m \leq size(b) \wedge \forall i. 0 \leq i < m \rightarrow b[i] > x$

(b)  $hasGreater(a, b) \equiv \forall 0 \leq i < size(a). \exists 0 \leq j < size(b). a[i] > b[j]$

(c)  $Extends(a, b) \equiv size(a) \leq size(b) \wedge \forall 0 \leq i < size(a). a[i] = b[i]$

8. (a) In general,  $e_1 = e_2$  does not logically imply  $e_1 \equiv e_2$ . A counterexample can be  $2 + 2 = 4$  but  $2 + 2 \not\equiv 4$ .

(b) In general,  $e_1 \neq e_2$  logically implies  $e_1 \not\equiv e_2$ . We know that syntactic equality implies semantic equality, or in other words,  $(e_1 \equiv e_2) \Rightarrow (e_1 = e_2)$ , the contrapositive of this statement is  $(e_1 \neq e_2) \Rightarrow (e_1 \not\equiv e_2)$ , which is also true.